Indian Institute of Technology, Jodhpur

Solution Sheet
Mathematics II
Problem Sheet I

Question 1

Let $G = \{0, 1, 2, 3\}$ with the binary operation $*$ defined by

$$a * b = (a + b) \mod 4.$$

(a) Verification that $(G, *)$ is a Group

To verify that $(G, *)$ is a group, we check the group axioms:

- **Closure:** For any $a, b \in G$, $a+b$ is an integer and $(a+b) \mod 4 \in \{0, 1, 2, 3\}$. Hence, $a * b \in G$. So, G is closed under $*$.
- **Associativity:** Addition modulo 4 is associative. For all $a, b, c \in G$,

  $$(a * b) * c = ((a + b) \mod 4 + c) \mod 4 = (a + (b + c)) \mod 4 = a * (b * c).$$

- **Identity Element:** The element $0 \in G$ satisfies

  $$a * 0 = (a + 0) \mod 4 = a = (0 + a) \mod 4 = 0 * a$$

  for all $a \in G$. Hence, 0 is the identity element.
- **Inverse Element:** For each $a \in G$, there exists an element $b \in G$ such that

  $$a * b = (a + b) \mod 4 = 0.$$

  Thus, every element has an inverse in G.

Therefore, $(G, *)$ is a group.

(b) Identity Element and Inverses

The identity element is 0.

The inverse of each element is given below:

Element a	Inverse of a
0	0
1	3
2	2
3	1

since

$$1 * 3 = 4 \mod 4 = 0, \quad 2 * 2 = 4 \mod 4 = 0.$$

Hence, each element of G has an inverse under the operation $*$.

Question 2

Try Yourself Using above definition.

Question 3

We show that the set of rational numbers $\mathbb{Q}$ forms a field under the usual addition and multiplication.

Recall that

$$\mathbb{Q} = \left\{ \frac{a}{b} \,\middle|\, a, b \in \mathbb{Z},\ b \neq 0 \right\}.$$

1. $(\mathbb{Q}, +)$ is an Abelian Group

- **Closure:** For any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$,

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \in \mathbb{Q}.$$

- **Associativity:** Addition of rational numbers is associative:

$$(x + y) + z = x + (y + z) \quad \text{for all } x, y, z \in \mathbb{Q}.$$

- **Additive Identity:** The element $0 = \frac{0}{1} \in \mathbb{Q}$ satisfies

$$x + 0 = x \quad \text{for all } x \in \mathbb{Q}.$$

- **Additive Inverse:** For any $\frac{a}{b} \in \mathbb{Q}$, the additive inverse is

$$-\frac{a}{b} = \frac{-a}{b} \in \mathbb{Q},$$

  and

$$\frac{a}{b} + \left(-\frac{a}{b}\right) = 0.$$

- **Commutativity:** Addition of rational numbers is commutative:

$$x + y = y + x \quad \text{for all } x, y \in \mathbb{Q}.$$

Hence, $(\mathbb{Q}, +)$ is an abelian group.

2. Multiplication Properties on $\mathbb{Q}$

- **Closure:** For any $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$,
$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}.$$
- **Associativity:** Multiplication of rational numbers is associative:
$$(xy)z = x(yz) \quad \text{for all } x, y, z \in \mathbb{Q}.$$
- **Multiplicative Identity:** The element $1 = \frac{1}{1} \in \mathbb{Q}$ satisfies
$$x \cdot 1 = x \quad \text{for all } x \in \mathbb{Q}.$$
- **Multiplicative Inverse:** For any nonzero rational number $\frac{a}{b} \in \mathbb{Q}$, the multiplicative inverse is
$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a} \in \mathbb{Q}, \quad a \neq 0.$$
- **Commutativity:** Multiplication of rational numbers is commutative:
$$xy = yx \quad \text{for all } x, y \in \mathbb{Q}.$$

3. Distributive Law

For all $x, y, z \in \mathbb{Q}$,
$$x(y + z) = xy + xz.$$

Since all the field axioms are satisfied, the set of rational numbers $\mathbb{Q}$ forms a field under the usual addition and multiplication.

Question 4

Let $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$ with addition and multiplication defined modulo 7.

Verification that $\mathbb{Z}_7$ is a Field

Since 7 is a prime number, every nonzero element of $\mathbb{Z}_7$ has a multiplicative inverse.

- $(\mathbb{Z}_7, +)$ forms an abelian group under addition modulo 7.
- Multiplication modulo 7 is associative and commutative.
- The additive identity is 0, and the multiplicative identity is 1.
- Every nonzero element in $\mathbb{Z}_7$ has a multiplicative inverse in $\mathbb{Z}_7$.
- Multiplication is distributive over addition.

Hence, $\mathbb{Z}_7$ is a field.

Multiplicative Inverse of 3 in $\mathbb{Z}_7$

We find an element $x \in \mathbb{Z}_7$ such that

$$3x \equiv 1 \pmod 7.$$

Checking values,

$$3 \times 5 = 15 \equiv 1 \pmod 7.$$

Therefore, the multiplicative inverse of 3 in $\mathbb{Z}_7$ is

$$3^{-1} \equiv 5 \pmod 7.$$

Question 5

Check yourself.

Question 6

We check whether V is a vector space over the field $F = \mathbb{R}$ in each case.

(i)

Given

$$V = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = 0\}, \quad F = \mathbb{R},$$

with vector addition and scalar multiplication defined as in $\mathbb{R}^2$.

We verify whether V is a vector space by checking the vector space axioms.

1. Non-emptiness and Zero Vector

The zero vector in $\mathbb{R}^2$ is $(0, 0)$. Since $x_2 = 0$, we have $(0, 0) \in V$. Thus, V is non-empty and contains the zero vector.

2. Closure under Addition

Let $(x_1, 0), (y_1, 0) \in V$. Then

$$(x_1, 0) + (y_1, 0) = (x_1 + y_1, 0 + 0) = (x_1 + y_1, 0).$$

Since the second component is zero, $(x_1 + y_1, 0) \in V$. Hence, V is closed under addition.

3. Closure under Scalar Multiplication

Let $(x_1, 0) \in V$ and $\alpha \in \mathbb{R}$. Then

$$\alpha(x_1, 0) = (\alpha x_1, \alpha \cdot 0) = (\alpha x_1, 0).$$

Since the second component is zero, $(\alpha x_1, 0) \in V$. Thus, V is closed under scalar multiplication.

4. Additive Inverse

For any $(x_1, 0) \in V$, its additive inverse is

$$-(x_1, 0) = (-x_1, 0),$$

which also belongs to V. Hence, every element of V has an additive inverse in V.

5. Other Axioms

Associativity and commutativity of addition, distributive laws, and compatibility of scalar multiplication all hold because $V \subseteq \mathbb{R}^2$ and the operations are inherited from $\mathbb{R}^2$.

$$\boxed{\text{True}}$$

(ii)

$$V = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}.$$

The zero vector $(0, 0) \notin V$ since $0 + 0 \neq 1$.
Hence, V is not a vector space.

$$\boxed{\text{False}}$$

(iii)

$$V = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 = 0\}.$$

Consider $(1, 0), (0, 1) \in V$. Then,

$$(1, 0) + (0, 1) = (1, 1) \notin V,$$

since $1 \cdot 1 \neq 0$.
Hence, V is not closed under addition.

$$\boxed{\text{False}}$$

(iv)

$$V = \{x \in \mathbb{R} : x \geq 0\},$$

with operations defined by

$$x + y := xy, \quad \alpha x := |\alpha| x.$$

The additive identity must satisfy $x + e = x \Rightarrow xe = x$, which implies $e = 1$.
However, the additive inverse of x must satisfy

$$x + y = 1 \Rightarrow xy = 1,$$

which does not exist in V for 0.
Hence, additive inverses do not exist.

$$\boxed{\text{False}}$$

(v)

Solution

Let

$$V = \mathbb{R}([a,b], \mathbb{R}),$$

the set of all real-valued Riemann integrable functions on the interval $[a, b]$, and let the field be $F = \mathbb{R}$. Vector addition and scalar multiplication are defined in the usual way:

$$(f+g)(x) = f(x) + g(x), \qquad (\alpha f)(x) = \alpha f(x),$$

for all $f, g \in V$, $\alpha \in \mathbb{R}$, and $x \in [a, b]$.

We verify the vector space axioms.

1. Non-emptiness and Zero Vector

The zero function $f(x) = 0$ for all $x \in [a, b]$ is Riemann integrable. Hence, the zero vector belongs to V.

2. Closure under Addition

Let $f, g \in V$. Since both f and g are Riemann integrable on $[a, b]$, their sum $f + g$ is also Riemann integrable on $[a, b]$. Thus,

$$f + g \in V.$$

3. Closure under Scalar Multiplication

Let $f \in V$ and $\alpha \in \mathbb{R}$. Then the function αf is Riemann integrable on $[a, b]$. Hence,

$$\alpha f \in V.$$

4. Additive Inverse

For any $f \in V$, the function $-f$ is also Riemann integrable, and

$$f + (-f) = 0.$$

Thus, every element of V has an additive inverse in V.

5. Other Vector Space Axioms

Associativity and commutativity of addition, distributive laws, and compatibility of scalar multiplication all hold since the operations are defined pointwise and inherit these properties from the real numbers.

$$\boxed{\text{True}}$$

(vi)

$$V = \{\text{all polynomials of degree 5 with real coefficients}\}.$$

The zero polynomial is not of degree 5. Hence, V does not contain the zero vector. Therefore, V is not a vector space.

$$\boxed{\text{False}}$$

(a) Using the Subspace Test

Let

$$V = P_3, \quad U = \{a_0 + a_1 t + a_2 t^2 + a_3 t^3 : a_0 = 0\}.$$

Subspace Test: A nonempty subset $U \subseteq V$ is a subspace if for all $p, q \in U$ and $\alpha \in \mathbb{R}$,

$$p + q \in U \quad \text{and} \quad \alpha p \in U.$$

Let

$$p(t) = a_1 t + a_2 t^2 + a_3 t^3, \quad q(t) = b_1 t + b_2 t^2 + b_3 t^3 \in U.$$

Then,

$$p(t) + q(t) = (a_1 + b_1)t + (a_2 + b_2)t^2 + (a_3 + b_3)t^3 \in U.$$

For any $\alpha \in \mathbb{R}$,

$$\alpha p(t) = \alpha a_1 t + \alpha a_2 t^2 + \alpha a_3 t^3 \in U.$$

Since U is nonempty (the zero polynomial belongs to U) and is closed under addition and scalar multiplication, U is a subspace of P_3.

$$\boxed{U \text{ is a subspace of } P_3}$$

(b)

$$V = P_3, \quad U = \{a_0 + a_1 t + a_2 t^2 + a_3 t^3 : a_2 = 0\}.$$

Then elements of U are of the form

$$p(t) = a_0 + a_1 t + a_3 t^3.$$

- The zero polynomial belongs to U.
- Sum of two elements in U is again in U.
- Scalar multiples remain in U.

Hence, U is a subspace of P_3.

$$\boxed{\text{Yes, } U \text{ is a subspace}}$$

(c)

$$V = \mathbb{R}^3, \quad U = \{\alpha v + \beta w : \alpha, \beta \in \mathbb{R}\},$$

where

$$v = (1, 4, 0), \quad w = (2, 2, 2).$$

The set U consists of all linear combinations of v and w.

- $0 = 0v + 0w \in U$.
- Closed under addition and scalar multiplication.

Hence, $U = \operatorname{span}\{v, w\}$, which is always a subspace of $\mathbb{R}^3$.

$$\boxed{\text{Yes, } U \text{ is a subspace}}$$